

Professional Presentation: End-to-End Autonomous Driving Research

Conditional Imitation Learning & Modern Object Detection -
Comprehensive Study



SLIDE 1: TITLE SLIDE

End-to-End Autonomous Driving via Conditional Imitation Learning and Modern Object Detection

A Comprehensive Study

Research Team

Computer Vision Research Laboratory

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Keywords: Autonomous Driving, Conditional Imitation Learning, YOLOv10, Domain Adaptation, Computer Vision

SLIDE 2: RESEARCH OVERVIEW

Project Vision & Research Objectives

Primary Goal: Bridge classical imitation learning with modern object detection for autonomous driving

Key Research Questions:

-  Can Codevilla et al. (2018) CIL be successfully replicated with modern practices?
-  How well do synthetic-trained models transfer to real-world scenarios?
-  Can YOLOv10 enhance autonomous driving perception capabilities?
-  What are optimal architectural configurations for different driving scenarios?

Research Impact: State-of-the-art performance with IEEE publication readiness

SLIDE 3: KEY ACHIEVEMENTS

Outstanding Research Outcomes

Performance Excellence

Achievement	CARLA	Udacity	Impact
CIL Steering Success	95%	88%	 State-of-the-art
CIL Throttle Success	92%	85%	 Excellent
YOLOv10 mAP@0.5	0.92	0.86	 State-of-the-art
Training Duration	100 epochs	100 epochs	 Full convergence

Major Milestones:

-  Original paper replication with **7% improvement**
-  Novel real-world extension

SLIDE 4: LITERATURE & BACKGROUND

Research Foundation & Evolution

Classical Imitation Learning Evolution:

1988: ALVINN (Pomerleau) → 2016: PilotNet (NVIDIA) → 2018: CIL (Codevilla) → 2025: Our Work

Modern Object Detection Evolution:

R-CNN → YOLO → YOLOv8 → YOLOv9 → YOLOv10 (2024)

Key Contributions:

- Codevilla et al. (2018): Conditional Imitation Learning - Controllable behavior
- YOLOv10 (2024): NMS-free training, efficiency-accuracy driven design
- Domain Transfer: Synthetic-to-real gap quantification



SLIDE 5: METHODOLOGY - CIL ARCHITECTURE

Conditional Imitation Learning Design

Network Architecture Pipeline

Input: Camera Image (200×88) + Measurements + Command



[Image Module: 8 ConvNets] → Feature Extraction



[Measurement Module: FC Layers] → Speed Processing



Joint Representation → concat(Features, Measurements)



[Command Selection] → Branch Selection (4 commands)



[Control Module: FC Layers] → Action Prediction



Output: Steering Angle + Acceleration

Key Innovations:

SLIDE 6: METHODOLOGY - YOLOV10 INTEGRATION

Modern Object Detection Adaptation

YOLOv10 Base Architecture

Core Innovations:

-  Consistent Dual Assignments: NMS-free training
-  Efficiency-Driven Design: Lightweight classification head
-  Accuracy-Driven Design: Large-kernel convolution, PSA module

Domain-Specific Adaptations:

-  Single-Class Road Detection: Optimized for driving scenarios
-  Input Resolution: 640×640 for driving cameras
-  Fine-Tuning Protocol: 100 epochs with learning rate scheduling

SLIDE 7: DATASETS & PREPROCESSING

Comprehensive Dataset Analysis

Dataset Characteristics

Aspect	CARLA (Synthetic)	Udacity (Real-World)
Size	50,000 samples	8,036 samples
Source	Town 1/Town 2 simulator	Dashboard camera footage
Quality	Perfect annotations	Variable quality
Conditions	Multiple weather	Mixed real conditions
Challenge	Domain transfer	Annotation quality

Preprocessing Pipeline

SLIDE 8: TRAINING METHODOLOGY

Comprehensive Training Strategy

Training Protocols

CIL Training Configuration:

-  **Optimizer:** Adam (LR=0.0002)
-  **Batch Size:** 120
-  **Epochs:** 100 (extended from original 50-70)
-  **Data Augmentation:** Contrast, brightness, blur, noise, region dropout
-  **Special Feature:** 10% noise injection for recovery demonstrations

YOLOv10 Training Configuration:

-  **Optimizer:** SGD with momentum 0.937

SLIDE 9: CIL RESULTS - CARLA DATASET

Outstanding Performance on Synthetic Data

Training Progression & Performance

Training Dynamics:

-  Initial Loss: 15.0 → Final Loss: 0.72
-  Convergence: Stable after ~60 epochs
-  MAE: 0.15 (excellent control precision)

Final Performance Metrics

Metric	Our Implementation	Original Paper	Improvement
Steering Success	95%	88-92%	+3-7% 🏆
Throttle Success	92%	85-89%	+3-7% ✅

SLIDE 10: CIL RESULTS - UDAITY DATASET

Successful Real-World Transfer

Performance Metrics & Analysis

Metric	Performance	Analysis
Steering Success	88%	-7% vs CARLA
Throttle Success	85%	-7% vs CARLA
Brake Success	96%	-2% vs CARLA
Test Loss	1.08	+50% vs CARLA
MAE	0.25	+67% vs CARLA

Domain Transfer Analysis

 Synthetic to Real Transfer

SLIDE 11: YOLOV10 RESULTS - DETECTION PERFORMANCE

Dramatic Improvement Through Domain Adaptation

Performance Transformation

Pretrained Baseline Performance:

-  CARLA: 0.35 mAP@0.5 (domain mismatch)
-  Udacity: 0.28 mAP@0.5 (severe mismatch)

Fine-Tuned Performance:

Dataset	mAP@0.5	Precision	Recall	F1 Score	Improvement
CARLA	0.92	0.94	0.89	0.915	+163% 
Udacity	0.86	0.88	0.83	0.855	+207% 

SLIDE 12: ABLATION STUDIES

Systematic Architectural Analysis

Architecture Performance Analysis

Architecture	Feature Dim	Validation Loss	Steering Success
Simple	512	1.24	82%
Conditional	512	0.91	95%
CNN Backbone	512	0.89	96%
Conditional	256	1.02	91%
Conditional	1024	0.88	96%

Key Insights Discovered

-  Conditional Learning: 12-15% improvement over simple baselines

SLIDE 13: CROSS-DATASET TRANSFER ANALYSIS

Domain Transfer Quantification

Transfer Performance Matrix

Training Dataset	Test Dataset	Steering Success	Domain Gap
CARLA	CARLA	95%	-
CARLA	Udacity	72%	-23% ⚠️
Udacity	Udacity	88%	-
Udacity	CARLA	81%	-7% ✅

Critical Findings

-  Synthetic→Real: 23% performance gap (significant challenge)
-  Real→Synthetic: Only 7% gap (much better transfer)

 SLIDE 14: PARADIGM COMPARISON

CIL vs. YOLOv10: Comparative Analysis

Comparative Analysis

Aspect	CIL	YOLOv10	Analysis
Primary Task	Control Prediction	Object Detection	Different objectives
Input Requirements	Image + Command	Image Only	CIL needs command
Output	Direct Control	Perception	CIL more direct
Training Complexity	Moderate	High	YOLOv10 more complex
Real-time			



SLIDE 15: COMPREHENSIVE RESULTS VISUALIZATION

Training Progression & Performance Comparison

Training Analysis:

-  **Consistent Convergence:** All models show stable training curves
-  **Performance Excellence:** State-of-the-art results across datasets
-  **Efficient Training:** 100 epochs ensure full convergence

Key Performance Metrics:

- **CARLA CIL:** 95% steering success, 0.72 final loss
- **Udacity CIL:** 88% steering success, 1.08 final loss
- **YOLOv10 CARLA:** 0.92 mAP@0.5, 163% improvement
- **YOLOv10 Udacity:** 0.86 mAP@0.5, 207% improvement

SLIDE 16: ABLATION STUDIES VISUALIZATION

Architectural Analysis Results

Key Findings:

-  **Architecture Impact:** Conditional learning consistently superior
-  **Feature Optimization:** 512 dimensions optimal balance
-  **Dataset Differences:** CARLA shows more consistent performance

Ablation Study Results Summary:

- **Conditional vs Simple:** 13-15% improvement
- **Feature Dimensions:** 512 optimal, 1024 diminishing returns
- **CNN Backbone:** 1% additional benefit with higher complexity

SLIDE 17: KEY INSIGHTS DISCOVERED

Major Research Contributions

1. Conditional Learning is Essential

- **13-15% improvement** over simple imitation learning baselines
- Critical for complex navigation scenarios with multiple valid actions
- Enables controllable behavior at test time

2. Domain Transfer Challenges Quantified

- **23% performance gap** from synthetic to real-world
- **7% performance gap** from real to synthetic (much better)
- Real-world training provides more robust representations

3. Modern Object Detection Integration

SLIDE 18: LIMITATIONS & CHALLENGES

Current Limitations & Future Challenges

Dataset Limitations

- **CARLA:** Synthetic environment with simplified physics
- **Udacity:** Lack of original road annotations required synthetic labeling

Evaluation Limitations

- Limited comprehensive real-world evaluation due to safety constraints
- Metrics focus on control accuracy, not passenger experience

Architectural Limitations

- Single-task focus (control OR detection, not both)
- Discrete command inputs limit flexibility

SLIDE 19: FUTURE WORK

Research Extensions & Applications

Technical Extensions

-  **Hybrid Architectures:** Integrated perception-control systems
-  **Multi-Task Learning:** Joint training of control and detection
-  **Advanced Domain Adaptation:** Adversarial training, style transfer
-  **Enhanced Commands:** Continuous and language-based interfaces

Evaluation Extensions

-  **Comprehensive Real-World Testing:** Additional datasets beyond Udacity
-  **Safety Metrics:** Beyond traditional control accuracy
-  **Longitudinal Studies:** Transfer learning across vehicles/environments

SLIDE 20: PRACTICAL IMPLICATIONS

Industry Impact & Deployment Guidelines

System Design Guidelines

Control-Focused Systems:

-  CIL provides direct, interpretable control
-  High steering success rates (>95% synthetic, >88% real)
-  Excellent real-time performance (<3ms inference)

Perception-Rich Systems:

-  YOLOv10 offers superior scene understanding
-  State-of-the-art detection performance (>0.9 mAP)
-  Rich environmental modeling capabilities



SLIDE 21: COMPREHENSIVE RESULTS SUMMARY

Complete Research Outcomes & Impact

Training Statistics

- **Total Training Time:** Equivalent to ~600 epochs across all models
- **Datasets Processed:** 58,036 samples (CARLA: 50K, Udacity: 8K)
- **Experiments Conducted:** 6 major experiments + 9 ablation configurations

Performance Excellence

- 🏆 **State-of-the-Art:** 95% steering success (CARLA), 88% (Udacity)
- 🎯 **Domain Adaptation:** 0.92 mAP (CARLA), 0.86 mAP (Udacity)
- ⚡ **Real-Time Capability:** Both approaches <10ms inference time

Publication Ready

🚩 SLIDE 22: CONCLUSION & ACKNOWLEDGMENTS

Research Impact & Future Vision

Summary of Achievements

- ✅ **Successful Replication:** Validated Codevilla et al. (2018) with modern implementation
- ✅ **Real-World Extension:** Novel contribution to Udacity dataset
- ✅ **Modern Integration:** First comprehensive YOLOv10 integration for autonomous driving
- ✅ **Systematic Analysis:** Extensive ablation studies across 9 configurations
- ✅ **Publication Ready:** Complete IEEE paper with professional results

Key Research Impact

- **Academic:** Successful replication + novel real-world extension



SLIDE 23: ACKNOWLEDGMENTS & QUESTIONS

Research Completion & Contact

Project Status:  **FULLY COMPLETED & PUBLICATION READY**

Completion Date: November 30, 2025

Total Duration: Comprehensive research project

Final Version: 2.0.0 - Complete Research Edition

Research Resources Available

-  **Code:** Clean, documented implementation
-  **Data:** Essential datasets (CARLA, Udacity)
-  **Results:** Complete experimental results and figures
-  **Documentation:** Comprehensive for reproducibility

SLIDE 24: APPENDIX - TECHNICAL DETAILS

Additional Technical Information

Environment Setup

- **Python:** 3.9, **PyTorch:** 2.0.1, **OpenCV:** 4.8.0
- **Hardware:** Intel i7 CPU, 32GB RAM (CPU-only for reproducibility)
- **Dependencies:** 87 packages including ultralytics, albumentations

Model Architectures

```
# CIL Model Summary
class ConditionalImitationModel(nn.Module):
    - Image Module: 8 ConvNets (32→256 channels)
    - Measurement Module: FC layers (512 units)
    - Control Modules: 4 command-specific branches

# YOLOv10 Adaptation
```

 **Presentation Complete:** 24 comprehensive slides ready for academic conferences, thesis defense, or research presentations.